Blessed G/Michael Catholic Secondary School

G-12 PHYSICS SECOND SEMESTER TEST 1 **DATE: MEGABIT 2018 EC**

TOPICS: **TIME ALLOWED: 1:20**

- **SIMPLE MACHINE & NUCLEAR PHYSICS**

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I. Write True if the statement is correct and False if it is incorrect.

T 1. Gamma radiation has no mass and no charge.

T 2. Radioactive isotopes are used in medicine.

T 3. Nuclear fission involves splitting a heavy nucleus.

T 4. Wheel barrow is an example of second order lever.

F 5. Machines reduce the amount of work needed to do a job.

F 6. After four half-lives, one-quarter of the initial mass remains.

T 7. A simple machine can multiply force or speed but not energy.

T 8. In simple machine, the work input is always greater than the work output.

F 9. The VR of a machine is three means the load moves three times the effort.

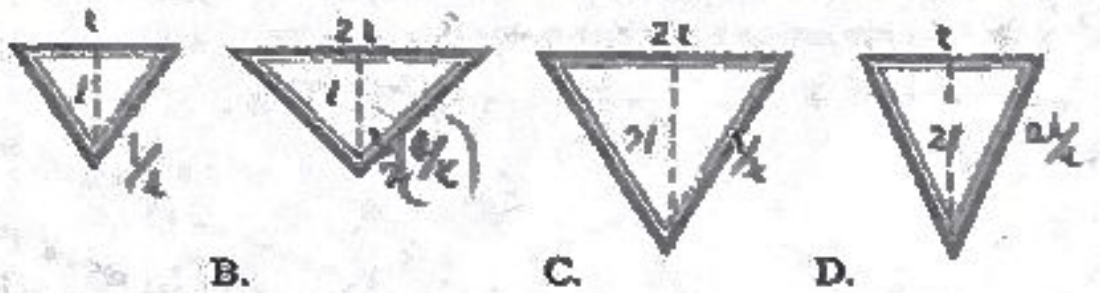
F 10. The strong nuclear force acts over long distances than electrostatic forces.

II. For each of the following questions, choose the best answer from the given alternatives.

B 11. Which can be stopped by paper?

A. Beta B. Alpha C. Gamma D. All of them

D 12. Which of the following double wedges with different length *l* and thickness *t* offers the greatest IMA?



A. B. C. D.

A 13. The activity of radioactive substance reduces from 200 Bq to 100 Bq in 10^6 seconds. What is the decay constant?

A. 6.93×10-7 decay /s C. 5 * 10 ^ - 7 decay /s

B. 6.93 * 10 ^ 5 decay /s D. 2 * 10 ^ 6 decay/s

D 14. Which of the following actions increases the efficienoy of a machine?

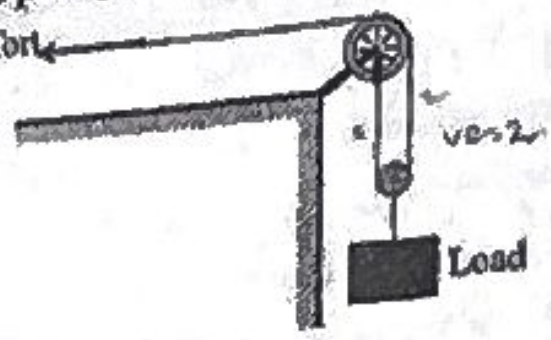
A. Decreasing the load. C. Decreasing distance moved by the load.

B. Decreasing the AMA. D. Decreasing the VR.

$\cos 30° = \sin 60° = \frac{\sqrt{3}}{2}, \cos 37° = \sin 53° = 0.8, \cos 45° = \sin 45° = \frac{\sqrt{2}}{2}, \cos 53° = \sin 37° = 0.6, \cos 60° = \sin 30° = 0.5$

15. Which of the following phenomena clearly demonstrates the particle nature of light?
   A. Refraction of light by prism
   B. Dispersion of white light
   C. The photoelectric effect
   D. Diffraction of light by double slit

16. The following figure represents Pulley system. If the pulley masses and friction losses are ignored, the machine is



   A. Speed multiplayer
   B. Force multiplayer
   C. Both Speed and Force multiplayer
   D. Neither speed nor force multiplayer

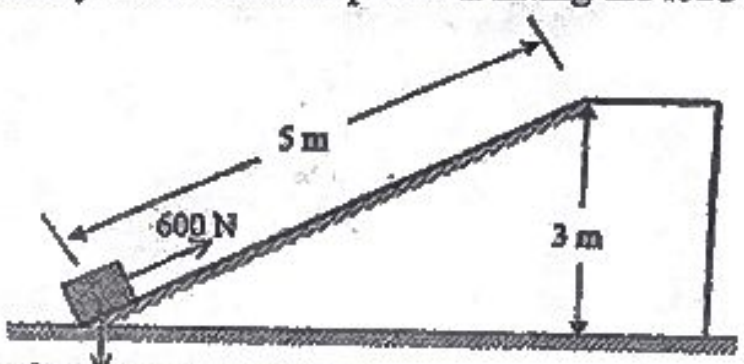
17. Which of the following is commonly used unit of nuclear energy?
   A. Joule
   B. Newton meter
   C. kilo calorie
   D. electron volt

18. A man pushes a 25.0 kg object by 300 N force along an inclined plane inclined at an angle of 37° to the horizontal. If the object moves with constant speed, the friction force exerts on the block is
   A. 150.0 N, up inclined plane.
   B. 100.0 N, down the inclined plane.
   C. 100.0 N, up the inclined plane.
   D. 150.0 N, down the inclined plane.

19. Which type of radiation has the greatest penetrating power?
   A. Alpha B. Beta C. Gamma D. All of them

20. A force that is exerted by a simple machine in order to produce an output work is called: __
   A. Normal B. effort C. load D. friction

21. Due to the presence of friction, the effort needed to slide a 600 N load along the inclined plane is 600 N as shown in the diagram below. What is the efficiency of the inclined plane in lifting the load to the 3 m height?
   A. 50 %
   B. 60 %
   C. 75 %
   D. 100 %



22. In photoelectric effect, the energy of photoelectrons is proportional to the:
   A. Speed of light
   B. Frequency of light
   C. Intensity of light
   D. Absorbance of the metal

23. When the radioactive element $^{144}Nd_{60}$ decays to $^{140}Ce_{58}$, which one of the following particles is emitted?
   A. beta particle B. proton C. alpha particle D. gamma ray

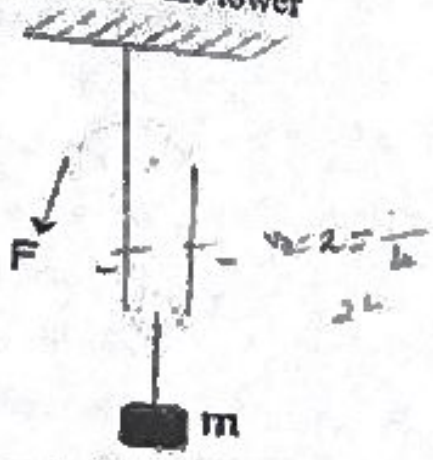
B 24. The following figure shows a simple machine that consists of a light, inextensible rope, light and frictionless pulleys, block of mass m, and applied force F. If the upper pulley is fixed and the lower one is movable, which of the following statements is NOT true?

A. If m raised vertically by h, then the rope where F is applied will move by a distance of 2h ✓

B. If m raised vertically by h, then the rope where F is applied will move by a distance of h. ✗

C. The mechanical advantage is 2 ✓

D. The velocity ratio is 2. ✓



B 25. The half-time of radium-224 is 3.5 days. What fraction of the sample remains undecayed after seven days?

A. ½ B. ¼ C. 1/8 D. 1/16

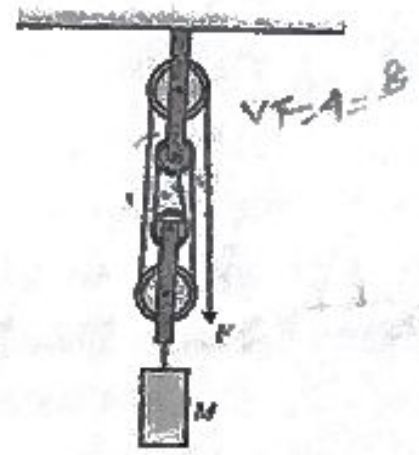
B 26. Consider the pulley system shown in the figure below. Which of the following is correct about the pulley system? (mass of pulleys and frictional forces are negligible)

A. Distances moved by the load and moved by the effort are equal ✗

B. If the effort moves 8cm, then the load will move 2cm ✓

C. The mechanical advantage of the system is $^1/_4$ ✗

D. The mechanical advantage of the system is 5 ✓



A 27. Electrons in a Sodium atom emit radiation when they move from a higher energy level of -0.81 eV to a lower energy level of -2.21 eV. What is the frequency of radiation emitted in the process?

A. $3.38 * 10^{14}$ Hz C. $8.10 * 10^{15}$ Hz

B. $1.40 * 10^{15}$ Hz D. $2.21 * 10^{14}$ Hz

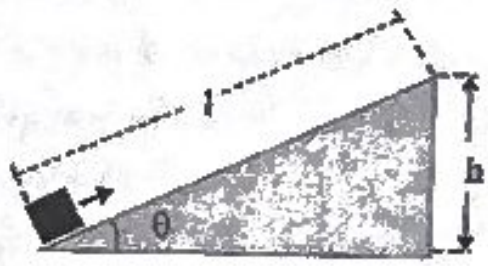
C 28. A force is used to push a box of mass, m along the slope of inclined plane with constant velocity as shown in the figure below. If the coefficient of friction between the box and the inclined plane be μ, the actual mechanical advantage (AMA) of the inclined plane is

A. $(\sin\theta + \mu\cos\theta)$

B. $(\cos\theta + \mu\sin\theta)$

C. $1/(\sin\theta + \mu\cos\theta)$

D. $1/(\cos\theta + \mu\sin\theta)$



29. Which one of the following statements describes the photoelectric effect interaction of radiation with matter?

A. Any value of energy of photons (radiations) B is capable of producing photoelectrons from a metal.

B. In photoelectric effect, the photoelectron C energy is directly proportional to the wavelength of incident radiation.

C. The intensity of radiation incident on a material can affect the energy of photoelectrons emitted from the metal surface.

D. When radiation of certain frequency f, interact with atoms of a material, electrons will be emitted with energies proportional to the radiation energy.

30. A 400 N force is required to be applied at one end of a 1m long lever, 75cm from its fulcrum, in order to lift a load of 600N placed at the other end as shown the figure below. What is the efficiency of the lever?

A. 25%

B. 50%

C. 67%

D. 75%



31. When ultraviolet light is shone on a clean metal surface the surface will emit electrons. What is the necessary precondition for electrons to be ejected from the metal surface is?

A. The intensity of light must be high enough

B. The minimum energy required to release B the electron is the same as the work function of the metal

C. The wave length of the light has to be greater than that of ultraviolet light.

D. The frequency of the metal has to be less than that of ultraviolet light.

32. A metallic wedge of thickness t = 1.5cm and length L = 9cm is used to split a wooden log (wood). If a force F_E = 80N applied on the wedge results to a load of F_L = 320N on the log, then which of the following statements is **not** correct about the wedge.

A. The velocity ratio is 6

B. The input work is 480J

C. The mechanical advantage (MA) of the wedge is 4

D. The efficiency of the wedge is 66.7%

33. The activity of a radioactive source falls to one sixteenth of its original value in 32 minutes. What is the half-life, in minutes, of this decay process?

A. 2 min B. 8 min C. 4 min D. 16 min

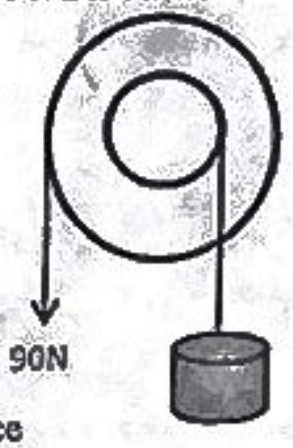
C 34. The diameters of the wheel and axle machine shown in the diagram below are 60 cm and 6 cm, respectively. A 90 N force applied to the rim of the wheel can lift a load of 810 N on the axle. The efficiency of the machine is

A. 51 %

B. 10 %

C. 90 %

D. 60 %



D 35. Which one is not the purpose of simple machine?

A. Change the direction of force
C. Multiply force
B. Multiply speed
D. Multiply energy

D 36. In Rutherford experiment, a beam of positively charge alpha particles was projected into a thin piece of gold foil. One of the observation of this experiment is that some charged particles deflected through a large angle (10° to 90°) are those concentrated in one place. What conclusion can be draw from this observation?

A. All the atom is positive charge

B. All the atom's negative charge

C. Most of the atom is empty space

D. Most of the mass and all positive charge is in tiny central nucleus

C 37. Complete the following equation which represents a nuclear fission reaction. What is the particle represented by X?

$$^{235}U + {}^{1}_{0}n \rightarrow {}^{95}Rb + {}^{139}Cs + 2X + \text{energy}$$

A. Electron
B. proton
C. neutron
D. alpha particle

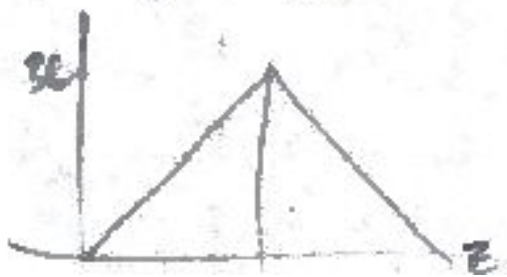
D 38. The graph of nuclear binding energy per nucleon (BE / A) versus mass number (A) has its peak near an atomic weight A = 56. Which of the following statements is true about the graph?

A. For smaller atoms (A < 56) the biding energy per nucleon generally increases with the increase of mass number; indicating that lighter atoms can increase stability through nuclear fission.

B. Uranium (A = 235) which has larger binding energy than iron (A = 56), is more stable than iron

C. Nucleus with larger number of neutron is more stable than nucleus of the same atom with smaller number of neutron

D. For heavier nuclei (A > 56), the binding energy per nucleon decreases with the increase of mass number; implying that heavier atoms can increase their stability by increasing their protons' number.



III. Short answer:

39. A radioactive substance has a half-life of 10 days.

a. What fraction remains after 20 days? 1/4 No

b. If the initial mass is 90g, how much remains after 30 days? 11.25g

40. Explain why the nucleus does not fly apart even though protons repel each other.

because there are neutrons in between protons which keep them separated with a strong nuclear force acting between the nucleons.

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1. True
2. True
3. True
4. True
5. False
6. False
7. True
8. True
9. False
10. False
11. B
12. D
13. A
14. D
15. C
16. B
17. BD
18. D
19. C
20. C
21. B
22. B
23. C
24. B
25. B
26. B
27. BA
28. C
29. D
30. B
31. B
32. B
33. B
34. C
35. D
36. D
37. C
38. D